ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ

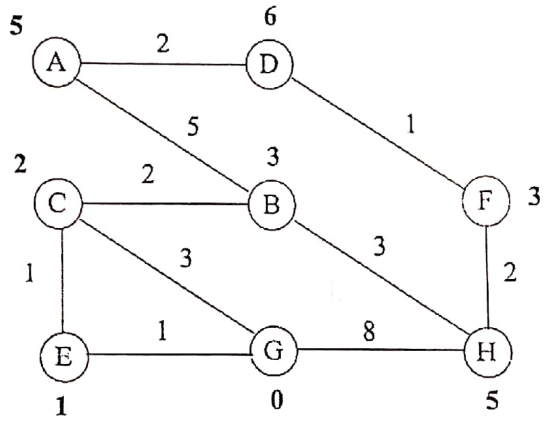
ĐỀ THI HỌC PHẦN TRÍ TUỆ NHÂN TẠO
HỌC KỲ II NĂM HỌC 2021 – 2022

Time: **75 minutes**

There are 2 pages and 4 questions in total. Students are allowed to use printed documents.

Question 1 (3 points).

Given a graph with the vertices and edges in the following figure. The edges' length is given by the number along with each edges. Each vertices is associated with an estimation distance to the Goal state, which is named by **G**. Assume if two nodes have the same priority, we choose to expand node with lower alphabetical order.



A → B → C → G (10)

a) Describe the steps for finding the path from **A** to **G** by **Bread First Search** algorithm. Write the path and its total distance.

b) Describe the steps for finding the path from **A** to **G** by **A*** algorithms. Write the path and its total distance.

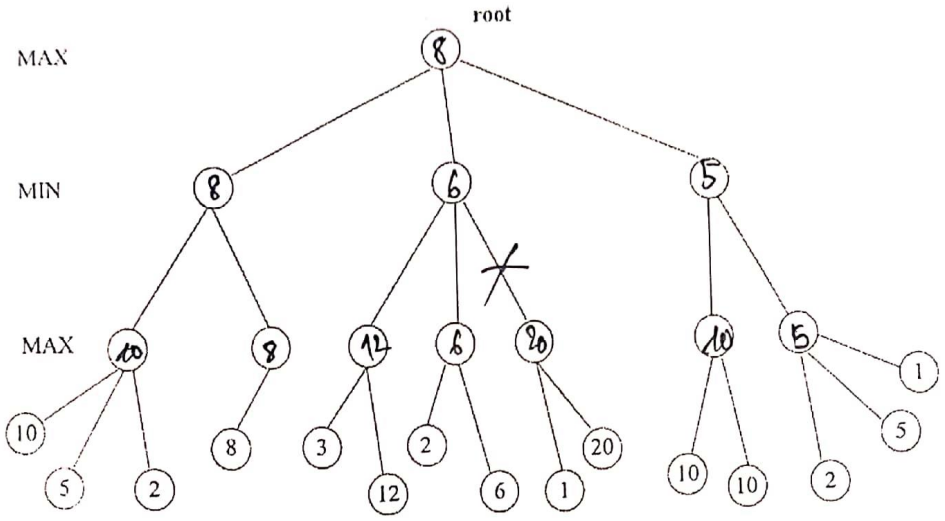
A → B → C → E → G (10)

Question 2 (3 points).

Given a minimax tree of two players as figure below. Find the value of the root node. Draw the path on the tree given both players play optimal path

a) Using minimax algorithm

b) Using alpha-beta pruning technique. Show the nodes that are pruned.



Question 3 (2 points).

Use First-Order-Logic to solve the following question:

Given:

- Giang and Long belong to the FIT club.
- Every member of the FIT club is either studying Machine Learning course or Image Processing course or both.
- No Machine Learning student likes chess, and all Image Processing student like selfie.
- Long dislikes whatever Giang likes and likes whatever Giang dislikes.
- Giang likes chess and selfie.

Query:

- Is there a member of the FIT club who is a Image Processing student but not a Machine Learning student?

Question 4 (2 points)

The students could choose one of two questions for answering:

a) Describe the Support Vector Machine (SVM) model. From your opinion, how to reduce the overfitting issues with SVM

b) What are some advantages of using Neural Network over a Perceptron (one Neural)? Why does training Deep Learning model need a lot of data?

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